

INTERNL: SCALING UP VISION FOUNDATION MODELS AND ALIGNING FOR GENERIC VISUAL-LINGUISTIC TASKS

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Open Source

MOTIVATION

There is a significant **parameter scale gap** between Large Language Models (LLMs) and Vision Foundation Models.

- **LLMs:** Typically scaled to **100B+** parameters.
- **Vision Models:** Stagnated at **~1B-3B** parameters.
- This disparity hinders the development of true Multi-modal AGI systems.

Goal: To bridge this gap by scaling up the vision encoder to **6B parameters** and progressively aligning it with LLMs.

ALIGNMENT STRATEGY

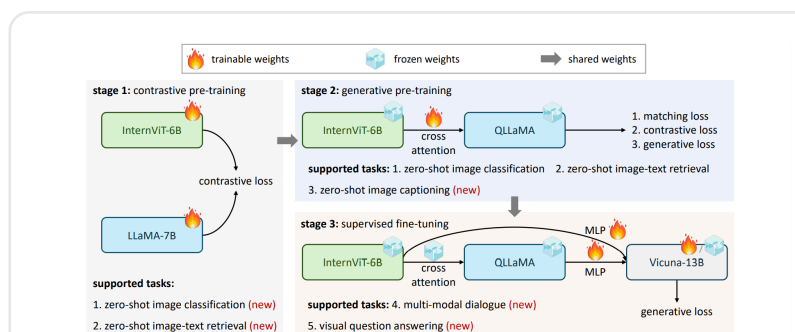


Figure 2: The 3-stage Progressive Alignment Strategy.

Training Process:

Stage 1: Contrastive Training

Vision-Language

Trained on massive noisy image-text pairs. Aligns InternViT with a text encoder.

Stage 2: Generative Training

V-L-LLM

Connects to QLLaMA. Focuses on high-quality data for captioning and grounding.

Stage 3: Supervised Fine-tuning

SFT

Fine-tuning on diverse datasets for state-of-the-art performance.

KEY RESULTS

Visual Perception

InternViT-6B achieves **90.1%** Top-1 accuracy on ImageNet-1K, surpassing previous SOTA vision encoders.

Multi-modal Tasks

Excellent performance on captioning, VQA, and retrieval tasks.

Flexibility

Can be used as a standalone vision encoder or integrated into LLMs to boost capabilities.

ARCHITECTURE

1. InternViT-6B

The largest open-source Vision Transformer (ViT) to date (6 billion parameters).

2. QLLaMA Bridge

A multi-modal alignment component based on Llama, connecting the vision encoder with the LLM.

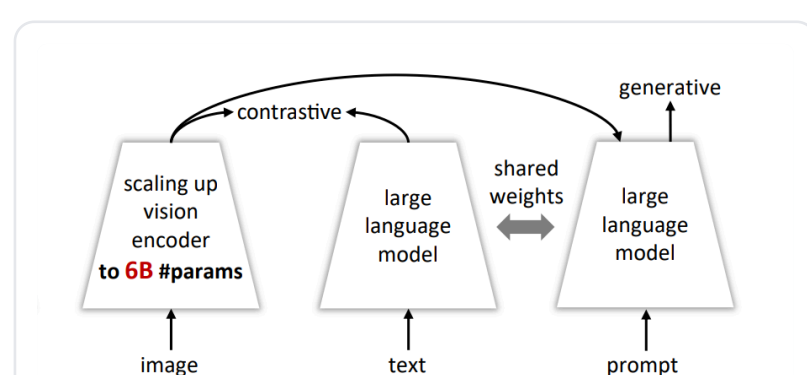


Figure 1: Overall Architecture of InternVL.

PERFORMANCE

Outperforming ViT-22B, BEiT-3, PALM-E.

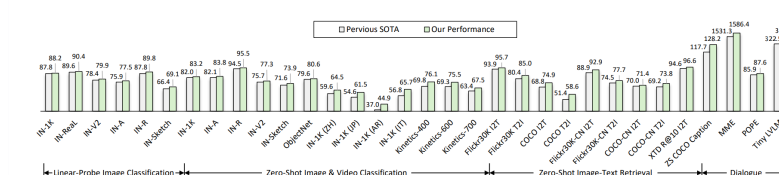


Figure 3: Performance Comparison.

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SOTA BENCHMARKS

Top-1

CLASSIFICATION

CONCLUSION

- **Bridging the Gap:** InternVL successfully scales vision models to 6B parameters.
- **Versatile Foundation:** A powerful visual encoder for Multi-modal systems.
- **Open Source:** Weights and code are fully available.

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SCAN ME

github.com/OpenGVLab/InternVL